

ZYFER

Zero-Trust sYnthEtic Forensics Engine Reinforcement

BENCHMARKING METHODOLOGY FOR MULTIMODAL DEEPPFAKE DETECTION PERFORMANCE

Prepared by: John McKenney
ZYFER Security Framework Initiative
Version 1.0 | April 2026

1. Purpose and Scope

This document establishes the formal benchmarking methodology for evaluating the detection performance of the ZYFER (Zero-Trust sYnthETic Forensics Engine Reinforcement) platform. It defines quantitative Key Performance Indicators (KPIs), scoring tiers, test conditions, and audit protocols across all four detection modalities that constitute ZYFER's multimodal threat analysis architecture.

The methodology is grounded in ZYFER's core design requirements as documented in the ZYFER Problem Statement and Solution Requirements specification, which establishes that the system must:

- Detect deepfake impersonations with high accuracy across facial biometrics, voice patterns, behavioral cues, and network metadata
- Operate in real time without disrupting communication flow, with a target inference latency at or below 50ms
- Apply without bias or discriminatory outcomes across demographic populations
- Provide explainable, auditable outputs to support human-in-the-loop oversight
- Align with NIST AI RMF 1.0 Measure and Manage functions, ISO/IEC 42001, and SOC 2 Type II

This methodology applies to all benchmark cycles: initial pilot deployment, post-optimization validation, organization-wide scale-up, and continuous operational monitoring.

2. ZYFER Detection Architecture: Benchmark Scope

ZYFER's detection pipeline operates across three architectural layers, each contributing distinct signals that are fused into a unified real-time threat confidence score. Benchmarking must be conducted independently at each layer and validated at the fusion layer to capture cross-modal consistency.

2.1 Data Ingestion Layer

Real-time capture from VoIP systems, enterprise video conferencing platforms, and cloud-based collaboration tools. Benchmarks at this layer measure stream integrity, latency at capture, and protocol handling.

2.2 Multimodal Detection Engine

The primary analytical layer. Benchmarks here are the most performance-critical and cover all four detection modalities:

CV	Computer Vision — Convolutional neural networks and vision transformers analyzing
----	---

	facial biometrics, micro-expression consistency, blink patterns, landmark geometry, and frame-level compression artifacts.
NLP	Speech & NLP — Audio forensics models evaluating spectral energy distribution, prosody, cadence, acoustic timbre, and conversational pacing for voice cloning and text-to-speech artifacts.
BEH	Behavioral Biometrics — Analysis of keystroke dynamics, interaction pacing, decision-making urgency, and unnatural automation signals within collaboration platforms.
NET	Network Metadata — Inspection of IP origin patterns, latency spikes, compression anomaly signatures, and spoofed connection indicators.

2.3 Correlation and Reinforcement Layer

Cross-modal fusion engine that synthesizes signals from all four modalities into a single threat confidence score. Benchmarks at this layer measure cross-modal consistency, fusion accuracy, and reinforcement learning adaptation rate under adversarial conditions.

3. KPI Framework

The following framework defines 15 KPIs across ZYFER's five benchmark domains. Each KPI specifies a baseline threshold (minimum acceptable performance), a target threshold (operational excellence standard), and a critical fail threshold (trigger for automated suspension and human review). KPIs are drawn directly from the ZYFER success criteria, technical requirements, and governance architecture.

3.1 KPI Domains

- **Detection Performance (Accuracy, FPR, FNR)**
- **Real-Time Operational Performance (Latency)**
- **Modality-Specific Performance (CV, NLP/Audio, Behavioral, Network)**
- **Fairness and Demographic Equity (Bias Delta Audits)**
- **Governance and Explainability (SHAP Coverage, HITL Accuracy)**

3.2 Master KPI Scorecard

Table 1 presents the complete ZYFER KPI scorecard. All benchmarks are evaluated per test cycle and reported as pass/audit/fail against defined thresholds.

KPI	Modality Layer	Baseline	Target	Critical Fail	Status
Detection Accuracy (Overall)	Multimodal	≥ 92%	≥ 97%	< 85%	PASS
False Positive Rate (FPR)	Multimodal	≤ 5%	≤ 2%	> 8%	PASS
False Negative Rate (FNR)	Multimodal	≤ 8%	≤ 3%	> 12%	PASS
Real-Time Inference Latency	Multimodal	≤ 100ms	≤ 50ms	> 200ms	PASS
Video Deepfake Detection Rate	Computer Vision	≥ 93%	≥ 98%	< 87%	PASS
Facial Landmark Consistency Score	Computer Vision	≥ 90%	≥ 96%	< 82%	PASS
Voice Deepfake Detection Rate	NLP / Audio	≥ 91%	≥ 97%	< 84%	PASS
Spectral Anomaly Detection Accuracy	NLP / Audio	≥ 89%	≥ 95%	< 80%	PASS
Behavioral Biometric Deviation Score	Behavioral Analytics	≥ 87%	≥ 94%	< 78%	PASS
Network Metadata Anomaly Detection	Network Analysis	≥ 88%	≥ 95%	< 79%	PASS
Cross-Modal Consistency Score	Multimodal Fusion	≥ 90%	≥ 96%	< 82%	PASS
Demographic Bias Delta (FPR)	Fairness Audit	≤ 3%	≤ 1%	> 5%	AUDIT

KPI	Modality Layer	Baseline	Target	Critical Fail	Status
Demographic Bias Delta (FNR)	Fairness Audit	≤ 3%	≤ 1%	> 5%	AUDIT
Model Explainability Coverage (SHAP)	Governance	≥ 95%	100%	< 90%	PASS
HITL Escalation Accuracy	Governance	≥ 90%	≥ 97%	< 80%	PASS

Table 1: ZYFER Multimodal Benchmarking KPI Scorecard

4. KPI Definitions and Measurement Protocols

4.1 Detection Performance KPIs

Detection Accuracy (Overall)

The percentage of total deepfake and authentic samples correctly classified across all four modalities in a fused inference pass. Computed as $(TP + TN) / \text{Total Samples}$. This is ZYFER's primary headline metric and must be evaluated across all three modality combinations: CV-only, NLP-only, and full-fusion.

False Positive Rate (FPR)

The rate at which authentic communications are incorrectly flagged as synthetic. Computed as $FP / (FP + TN)$. FPR is a critical operational KPI in ZYFER because false positives directly disrupt legitimate executive and employee communications — the exact failure mode described in ZYFER's ethical risk scenario of a call being dropped for a user experiencing network degradation or poor lighting.

NOTE

FPR must be evaluated separately for each demographic cohort (race, gender, age group, accent profile) as a fairness prerequisite. Aggregate FPR within threshold does not satisfy the fairness requirement if cohort-level FPR deltas exceed the defined bias tolerance.

False Negative Rate (FNR)

The rate at which synthetic impersonations are not detected. Computed as $FN / (FN + TP)$. FNR represents direct security failure — an undetected deepfake attack reaching the target executive or employee. This metric is weighted as the highest-severity failure state in ZYFER's operational model.

4.2 Real-Time Operational Performance

Real-Time Inference Latency

End-to-end processing time from raw stream capture at the Data Ingestion Layer to threat confidence score delivery at the user interface, measured in milliseconds. ZYFER targets sub-50ms latency to ensure detection occurs faster than human cognitive processing of a live communication. Measurements must be taken under peak load conditions (concurrent sessions) using cloud-compute configurations across AWS, Azure, and GCP to validate cloud-agnostic performance.

TARGET

≤ 50ms end-to-end latency at 95th percentile under enterprise load. This corresponds to ZYFER's documented specification and enables real-time interruption of synthetic impersonation before any sensitive action can be authorized.

4.3 Modality-Specific Performance KPIs

Video Deepfake Detection Rate

Accuracy of the Computer Vision module in classifying synthetic video frames. Test sets must include: face-swap deepfakes, GAN-generated avatars, lip-sync injection attacks, and live video with known degradation conditions (compression artifacts, low bandwidth, poor lighting). Evaluated using CNN and vision transformer model variants as documented in the ZYFER technical specification citing the Javed et al. (2025) transformer framework for gaze and blink pattern analysis.

Facial Landmark Consistency Score

A composite score measuring temporal consistency of detected facial landmarks across sequential video frames. Inconsistencies in landmark geometry, reflection patterns, and micro-expression timing are primary deepfake indicators documented in ZYFER's AI technology selection rationale. Scored on a normalized 0–100 scale; threshold values represent minimum acceptable consistency for authentic classification.

Voice Deepfake Detection Rate

Accuracy of the NLP and audio forensics module in detecting voice cloning, text-to-speech generation, and voice conversion attacks. Test sets must include samples across a broad range of acoustic profiles — accent variation, age group, gender expression, and emotional state — to capture the spectral energy distribution and prosodic irregularities documented in ZYFER's architecture as primary voice deepfake markers. Evaluation benchmarked against the Zhang et al. (2025) framework for audio deepfake detection.

Spectral Anomaly Detection Accuracy

Accuracy of the audio forensics module in identifying AI-generated artifacts in the spectral domain, including unnatural frequency distributions, cadence breaks, and acoustic timbre inconsistencies. Measured independently from conversational NLP scoring to isolate signal-level anomalies from linguistic manipulation patterns.

Behavioral Biometric Deviation Score

Accuracy of the behavioral analytics module in detecting synthetic interaction patterns, including: automated keystroke injection, unnatural mouse movement curves, atypical decision-making urgency patterns, and social engineering linguistic cues (manufactured urgency, abnormal phrasing, authority exploitation). Benchmarked using both automated agent simulations and expert human social engineering scenarios.

Network Metadata Anomaly Detection Rate

Accuracy of the network analysis module in flagging suspicious connection characteristics: spoofed IP origins, latency spike signatures consistent with synthetic media streaming, compression artifact patterns, and abnormal packet structures. Evaluated against a curated dataset of known deepfake delivery infrastructure signatures.

Cross-Modal Consistency Score

A composite metric measuring alignment between all four modality signals within a single detection event. High-confidence authentic sessions should produce consistent signal patterns across CV, NLP, behavioral, and network layers. Inconsistency between modalities — for example, a face passing CV analysis but failing voice spectral analysis — elevates the threat confidence score and triggers HITL escalation. This metric validates the fundamental ZYFER thesis that no single modality is sufficient for reliable detection.

4.4 Fairness and Demographic Equity KPIs

Demographic Bias Delta — FPR and FNR

The maximum observed difference in False Positive Rate or False Negative Rate between any two demographic cohorts within a test population. Cohorts are defined across race, gender, age group, accent profile, and skin tone classifications — the demographic axes historically associated with performance disparities in facial recognition and voice analysis systems.

Measurement uses SHAP and LIME analysis to attribute model decisions to specific input features, revealing differential behavior across cohorts. Statistical parity is evaluated using demographic parity difference and equalized odds metrics. Any bias delta exceeding the defined threshold triggers an immediate AUDIT flag, suspending deployment for the affected cohort pending retraining.

CRITICAL	Demographic Bias Delta is not a secondary metric — it is a prerequisite for deployment certification. ZYFER cannot operate in a production environment where bias delta exceeds the defined threshold for any cohort, regardless of aggregate detection accuracy. This directly addresses the ZYFER governance requirement that security enhancements do not come at the expense of civil liberties or fairness.
-----------------	--

4.5 Governance and Explainability KPIs

Model Explainability Coverage (SHAP)

The percentage of AI-driven detection decisions for which SHAP (SHapley Additive exPlanations) value attribution is successfully generated and logged in the audit trail. A coverage rate below 100% indicates gaps in the explainability layer that may prevent authorized personnel from reviewing the reasoning behind specific detection outcomes. Target is 100% coverage; critical fail is below 90%.

HITL Escalation Accuracy

The percentage of cases escalated to human review by ZYFER's Human-in-the-Loop mechanism that are subsequently confirmed as correctly classified threats by the reviewing security analyst. This metric validates that the escalation thresholds are calibrated correctly — low HITL accuracy indicates the system is escalating noise rather than genuine ambiguous cases, degrading analyst trust and response capacity.

5. Scoring Rubric and Alert Protocol

All KPI results are evaluated against the five-tier scoring rubric defined below. Tier classification triggers specific operational responses to ensure that detection quality degradation is addressed before it reaches user-facing impact.

Score Tier	Alert Color	Action Protocol
Exceeds Target	Green	KPI meets or exceeds target threshold across all demographic cohorts. No corrective action required.
Meets Baseline	Yellow	KPI meets baseline but has not reached target. Optimization recommended within 30 days.
Below Baseline	Orange	KPI falls between baseline and critical fail threshold. Escalate to model retraining team. HITL mandatory.
Critical Fail	Red	KPI breaches critical threshold. Automated actions suspended. Immediate human override required. Incident report filed.
Audit Required	Blue	Demographic bias delta exceeds tolerance. Full SHAP/LIME audit triggered. Deployment paused for affected cohort.

Table 2: ZYFER Benchmarking Scoring Rubric and Alert Protocol

6. Test Conditions and Dataset Requirements

6.1 Dataset Composition

Benchmark test datasets must meet the following composition requirements to ensure valid and representative evaluation:

- Minimum 10,000 labeled samples per modality (video, audio, behavioral, network)
- Minimum 40% synthetic/deepfake samples across all major generation methods: GANs, diffusion models, voice conversion, face-swap, lip-sync injection
- Demographic representation achieving statistical parity across race, gender, age group (18–25, 26–45, 46–65, 65+), and accent profile cohorts
- Inclusion of adversarial edge cases: low-bandwidth streams, compression artifacts, poor lighting, background noise, non-native speech patterns
- Sourced from voluntary contributors, institutional research partnerships, and publicly available deepfake detection challenge datasets (FaceForensics++, ASVspoof, DeepFakeTIMIT, DFDC)
- Updated quarterly to incorporate samples generated by newly released deepfake models

6.2 Test Environment Specifications

All benchmark cycles are executed across three cloud environments (AWS, Azure, GCP) to validate the cloud-agnostic performance claim. Test conditions must include:

- Simulated peak enterprise load (concurrent session counts representative of organizational scale)
- Variable network conditions: nominal bandwidth, degraded bandwidth (simulating legitimate compression artifacts), and controlled adversarial stream injection
- Cold-start and warm-model latency measurements to distinguish edge inference from cloud-resident inference
- Isolation of each modality module for independent scoring prior to fusion-layer evaluation

6.3 Adversarial Test Protocol

ZYFER's reinforcement learning architecture requires benchmarking under adversarial conditions to validate model robustness against data poisoning and model evasion — two primary attack vectors documented in ZYFER's threat model. Adversarial benchmarks must include:

- Data poisoning resistance: injection of mislabeled synthetic samples into test batch; measure detection accuracy degradation
- Model evasion resistance: GAN-generated samples specifically optimized to defeat ZYFER's published detection patterns
- Cross-modal spoofing: authentic video paired with synthetic audio, and vice versa, to test fusion-layer disambiguation
- Temporal consistency attacks: deepfake video with artificially corrected landmark geometry to defeat CV-layer analysis

7. Benchmark Cycle Schedule

ZYFER benchmarking is integrated into the three-phase implementation roadmap defined in the ZYFER Implementation Strategy, with ongoing post-deployment monitoring cycles.

Phase	Name	Timeline	Benchmarking Activities
Phase 1	Pilot Deployment	Weeks 1–4	Deploy to IT security staff + select executives. Establish baselines for all 15 KPIs. Capture false alarm rates and latency under live conditions.
Phase 2	Feedback & Optimization	Weeks 5–10	Retrain models on pilot data. Optimize threshold tuning. Run SHAP/LIME fairness audits across demographic segments. Reduce FPR to target tier.
Phase 3	Organization-Wide Scale-Up	Weeks 11–20	Full enterprise rollout. Activate HITL escalation protocols. Enable automated risk reporting. Benchmark cross-modal consistency at scale.
Ongoing	Continuous Monitoring	Post Week 20	Model drift detection, adversarial retraining cycles, quarterly bias audits, SIEM log integration, and annual governance review.

Table 3: ZYFER Benchmark Cycle Schedule Aligned to Implementation Phases

8. Governance Framework Alignment

This benchmarking methodology is designed to satisfy the Measure and Manage functions of the NIST AI Risk Management Framework (AI RMF 1.0) and the audit requirements of ISO/IEC 42001:2023, as specified in ZYFER's governance architecture.

Governance Requirement	Framework	ZYFER KPI Mapping
Quantitative risk measurement of AI system performance	NIST AI RMF – Measure	All 15 KPIs; baseline and target thresholds
Risk treatment and model improvement triggers	NIST AI RMF – Manage	Scoring rubric tiers and alert protocols
Documented AI policy objectives and audit trails	ISO/IEC 42001	SHAP Coverage KPI; HITL Escalation Accuracy
Demographic fairness and equitable performance	NIST AI RMF; ISO/IEC 42001	Demographic Bias Delta KPIs (FPR & FNR)
Human oversight for high-stakes automated decisions	HITL Policy; NIST Govern	HITL Escalation Accuracy KPI; Critical Fail protocol
Continuous improvement	ISO/IEC 42001; NIST	Adversarial test protocol; quarterly dataset

Governance Requirement	Framework	ZYFER KPI Mapping
and model retraining cycles	Manage	refresh
Evidence of SOC 2 Type II operational controls	SOC 2 Type II	Audit log completeness; SHAP Coverage KPI

Table 4: Governance Framework Alignment Matrix

9. Reporting and Output Artifacts

Each benchmark cycle produces the following standardized reporting artifacts, which are logged to ZYFER's audit trail and synchronized to the organization's SIEM platform:

- **KPI Scorecard Report** — Full 15-KPI scorecard with tier classifications, trend deltas from prior cycle, and cohort-level breakdowns for all fairness KPIs
- **SHAP Attribution Summary** — Feature-level contribution analysis for a stratified random sample of detection decisions, formatted for security analyst review
- **Adversarial Test Report** — Results of the four adversarial test protocols with model resilience scores and recommended retraining priorities
- **Bias Audit Certificate** — Demographic cohort performance analysis; must be countersigned by designated AI Governance Officer before next deployment phase is authorized
- **HITL Escalation Log** — Complete record of all human review escalations with analyst outcomes, used to recalibrate escalation confidence thresholds

10. Future Benchmark Evolution

This methodology is a living document. As ZYFER's detection capabilities expand and as deepfake generation technologies evolve, benchmark KPIs, thresholds, and test protocols will be updated through a formal versioning process. Planned extensions include:

- GAN-version-specific detection benchmarks as new generative models are released
- Real-time behavioral benchmark expansion to cover synthetic co-pilot and AI-assisted participant detection in collaboration tools
- Federated learning performance benchmarks for privacy-preserving distributed model updates
- Regulatory benchmark mapping for sector-specific standards in financial services (FFIEC), healthcare (HIPAA AI guidance), and government (CISA AI cybersecurity framework)
- Integration of TagUI-based automated benchmark orchestration for continuous regression testing across modality modules

References

Javed, M., Zhang, Z., Dahri, F.H., Laghari, A.A., Krajčík, M. et al. (2025). Real-Time Deepfake Detection via Gaze and Blink Patterns: A Transformer Framework. *Computers, Materials & Continua*, 85(1), 1457–1493. <https://doi.org/10.32604/cmc.2025.062954>

Zhang, B., Cui, H., Nguyen, V., & Whitty, M. (2025). Audio deepfake detection: What has been achieved and what lies ahead. *Sensors*, 25(7), 1989. <https://doi.org/10.3390/s25071989>

Tabassi, E. (2023). Artificial intelligence risk management framework (AI RMF 1.0). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>

Chu, C. H., Donato-Woodger, S., Khan, S. S., et al. (2023). Age-related bias and artificial intelligence: a scoping review. *Humanities and Social Sciences Communications*, 10(1). <https://doi.org/10.1057/s41599-023-01999-y>

McKenney, J. (2025). ZYFER – Zero-Trust sYnthEtic Forensics Engine Reinforcement: Problem Statement and Solution Requirements. ZYFER Security Framework Initiative.

McKenney, J. (2026). ZYFER Documentation – Zero-Trust Synthetic Forensics Engine. https://jmckenney3563.github.io/ZYFER_WEBSITE/docs.html